SMOOTHNESS AFTER CUTTING BY A QUASI-REGULAR SEQUENCE OVER A LOCALLY NILPOTENT IDEAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Flatness of the cut algebra along a quasi-regular sequence. Lemma \ref{lemma-cut-by-koszul-flat}, cited at proof lines 8076-8077. Gives flatness of the cut map.
- Smoothness is equivalent to flatness plus regular fibres. Algebra, Lemma \ref{algebra-lemma-flat-fibre-smooth}, cited at proof lines 8080-8082. Turns flatness and smooth fibres into smoothness.

Source reference. Original lemma 0CER: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15586–15586. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0CEQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15585–15585. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00TF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1073–1073. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $A' \to B'$ be a ring map. Let $I \subset A'$ be an ideal. Set $A = A'/I$ and $B = B'/IB'$. Let $f'_1, \ldots, f'_r \in B'$. Assume*

(1) *$A' \to B'$ is flat and of finite presentation (for example smooth),*

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(2) *I is locally nilpotent,*
(3) *the images $f_1, \ldots, f_r \in B$ form a quasi-regular sequence,*
(4) *$B/(f_1, \ldots, f_r)$ is smooth over A.*

Then $B'/(f'_1, \ldots, f'_r)$ is smooth over A'.

Proof. Set $C' = B'/(f'_1, \ldots, f'_r)$ and $C = B/(f_1, \ldots, f_r)$. Then $A' \to C'$ is of finite presentation. By Lemma 0CEQ we see that $A' \to C'$ is flat. The fibre rings of $A' \to C'$ are equal to the fibre rings of $A \to C$ and hence smooth by assumption (4). It follows that $A' \to C'$ is smooth by Algebra, Lemma 00TF. □